



Lecture 4: Underground Cables

EEE 321: Power System I

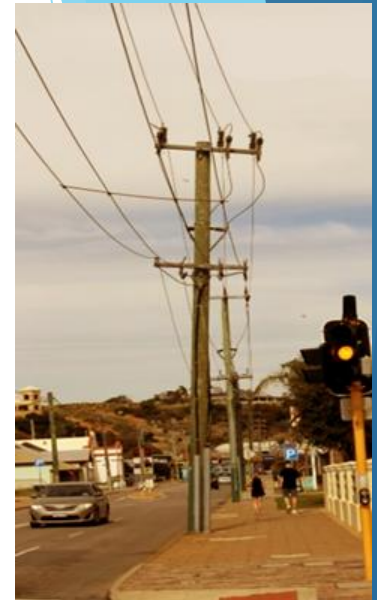
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Presentation Outline

- ❑ Comparison between Overhead lines and Underground Cables
- ❑ Construction
- ❑ Insulation Properties
- ❑ Insulation Resistance
- ❑ Problems
- ❑ Cables for 3-Phase service

Comparison between Overhead lines and Underground Cables

Sr. No.	Underground system	Overhead system
1.	Transmission is by using the cables.	Transmission is by using the transmission lines.
2.	All the cables must be properly insulated from each other.	The appropriate spacing provided between the conductors acts as an insulation. No external insulation is necessary.
3.	The insulation cost is very high.	No insulation cost as air acts as an insulator.
4.	Transmission over long distance is not possible as laying of cables is difficult, costly and complicated.	Transmission over long distance is possible with the help of transmission lines.
5.	The voltage level used is below 66 kV due to insulation difficulties.	The voltage level used can be as high as 400 kV.
6.	The maintenance cost is less.	The maintenance cost is high.
7.	The faults due to lightening, short circuit, storms etc. are eliminated.	The occurrence of faults due to lightening, short circuits, storms and abnormal weather conditions are possible.
8.	It is very safe due to insulation used.	The conductors are bare without insulation hence dangerous.
9.	Maximum stress is on the insulation between the conductors.	Maximum stress is between conductor and earth.
10.	The beauty of the area, towns etc. is well maintained.	The beauty of the area gets affected due to overhead lines. Sometimes trees are required to be cut.
11.	The size of the cables is high.	The size of the conductors is less.
12.	The voltage drop is less.	The voltage drop is more.

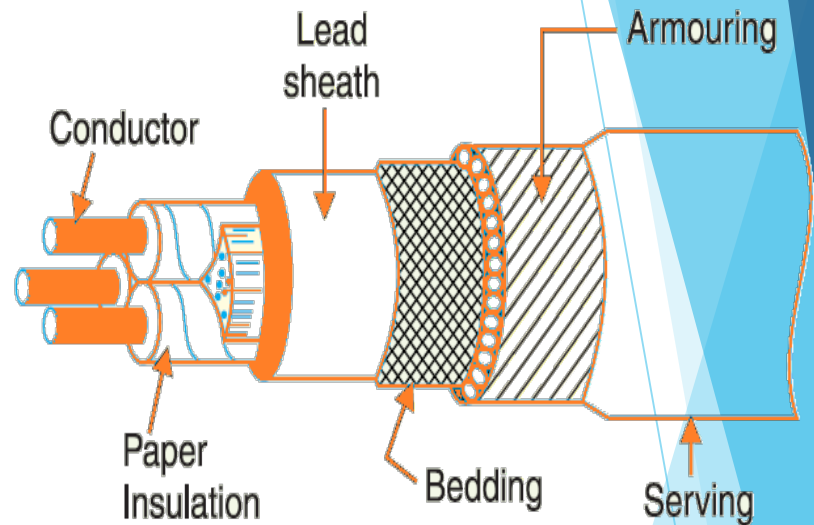


Underground Cable Construction

(i) Conductors: One or more than one conductor (**core**) depending upon the system. The conductors are made of copper or aluminium and are usually stranded in order to provide flexibility to the cable.

(ii) Insulation: Each conductor is provided with an insulation of suitable thickness so as to avoid leakage of current. The thickness of insulation layer depends upon the voltage to be withstood by the cable. The commonly used materials for insulation are impregnated paper, varnished cambric or rubber mineral compound.

(iii) Metallic sheath: In order to protect the cable from moisture, gases or other damaging liquids (acids or alkalies) in the soil and, a metallic sheath of lead or aluminium is provided over the insulation.

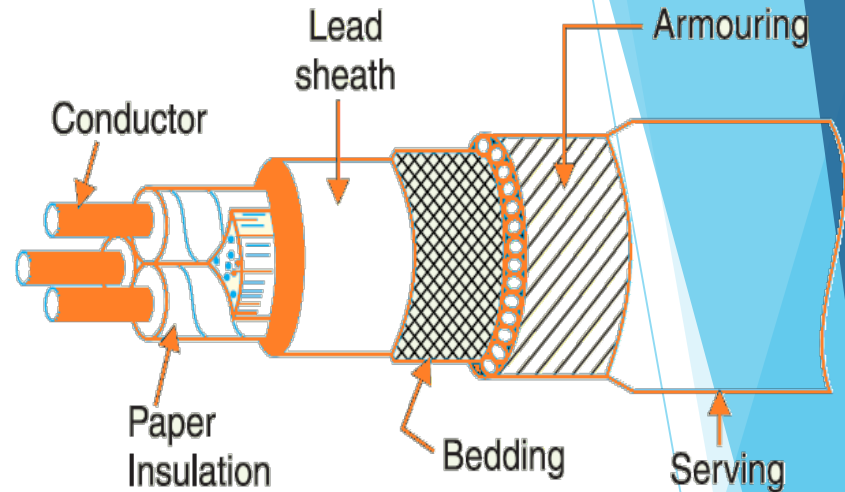


Underground Cable Construction

(iv) Bedding. Over the metallic sheath is applied a layer of bedding which consists of a fibrous material like jute or hessian tape. The purpose of bedding is to protect the metallic sheath against corrosion and from mechanical injury due to armouring.

(v) Armouring. Over the bedding, armouring is provided which consists of one or two layers of galvanised steel wire or steel tape. Its purpose is to protect the cable from mechanical injury while laying it and during the course of handling.

(vi) Serving. In order to protect armouring from atmospheric conditions, a layer of fibrous material (like jute) similar to bedding is provided over the armouring. This is known as serving.



Bedding, armouring and serving are only applied to the cables for the protection of conductor insulation and to protect the metallic sheath from mechanical injury.

Insulation Properties of Underground Cables

In general, insulating material used inside underground cables should have the following properties:

- (i) High insulation resistance to avoid leakage current.
- (ii) High dielectric strength to avoid electrical breakdown of the cable.
- (iii) High mechanical strength to withstand the mechanical handling of cables.
- (iv) Non-hygroscopic *i.e.*, it should not absorb moisture from air or soil. The moisture tends to decrease the insulation resistance and hastens the breakdown of the cable. In case the insulating material is hygroscopic, it must be enclosed in a waterproof covering like lead sheath.
- (v) Non-inflammable.
- (vi) Low cost so as to make the underground system a viable proposition.
- (vii) Unaffected by acids and alkalies to avoid any chemical action.

No one insulating material possesses all the above mentioned properties. Therefore, the type of insulating material to be used depends upon the purpose for which the cable is required and the quality of insulation to be aimed at. The principal insulating materials used in cables are rubber, vulcanised India rubber, impregnated paper, varnished cambric and polyvinyl chloride.

Insulation Properties of Underground Cables

Three main types of insulation

- **Paper-Oil insulation**

- High-pressure, fluid-filled pipe (**HPFF**)
- Self-contained fluid-filled (**SCFF**)

- **Paper-Gas insulation**

- High-pressure, gas-filled pipe (**HPGF**)

- **Plastic insulation**

- Solid cable, cross-linked polyethylene (**XLPE**)



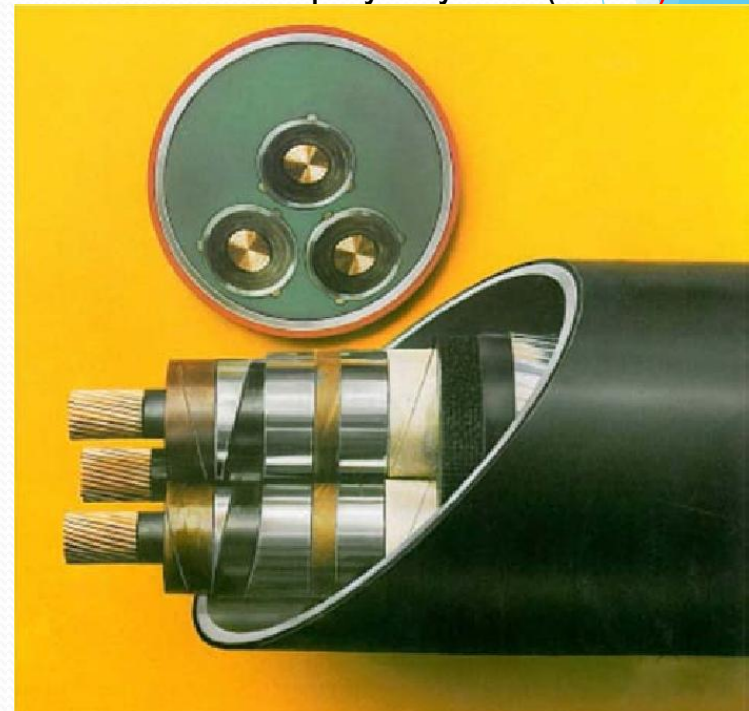
cross-linked polyethylene (**XLPE**)



High Pressure Fluid-Filled Pipe-Type Cables (**HPFF**)



Self Contained Fluid-Filled (**SCFF**) Cables



Gas- Filled Cables

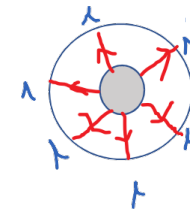
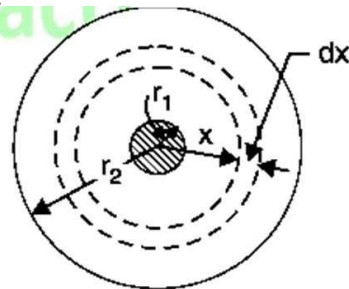
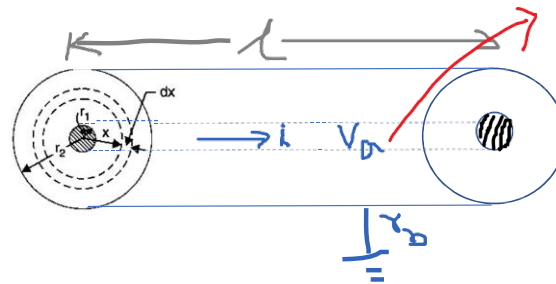
Insulation Resistance of a Single-Core Cable

Consider a single-core cable of conductor radius r_1 and l be the length of the cable. So, current i flows through this conductor or core and it has a potential V_a .

Outside of this conductor is surrounded by an insulating material. This insulating material is connected to the ground which has a potential also, V_b .

Due to the potential difference between conductor and ground ($V_a - V_b$), there is a small leakage current flowing from conductor to ground.

In order to prevent leakage current flowing, the cable conductor is provided with a suitable thickness of insulating material. For satisfactory operation, the insulating material/insulation resistance of the cable should be very high.



The path for leakage current is radially outwards

Insulation Resistance of a Single-Core Cable

Consider

r_1 = Radius of the conductor

r_2 = Radius of the cable = Radius of Internal insulating sheath of the cable

t = Thickness of the insulating material = $r_2 - r_1$

l = Length of the cable and

ρ = Resistivity of the insulation

From the total thickness of insulation, consider a very small layer of insulation of thickness dx at a distance x from the center of the cable.

So when thickness is dx , then leakage current tends to flow through the thickness (length) of insulation is dx and the area where the leakage current offer to flow is $2\pi x l$ (A cylinder of length l and radius x). [Cross section area = circumference for flowing leakage current multiplied by length of cable = $(2\pi x) \times l$]

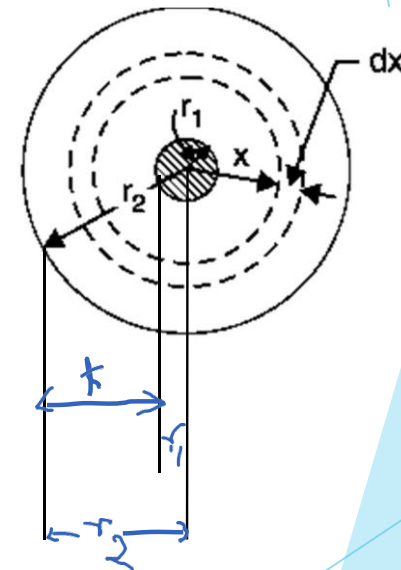
$$\therefore \text{Insulation resistance of considered layer} = \rho \frac{dx}{2\pi x l}$$

Insulation resistance of the whole cable is

$$R = \int_{r_1}^{r_2} \rho \frac{dx}{2\pi x l} = \frac{\rho}{2\pi l} \int_{r_1}^{r_2} \frac{1}{x} dx$$

$$R = \frac{\rho}{2\pi l} \ln \frac{r_2}{r_1}$$

Insulation resistance of a cable is inversely proportional to its length. In other words, if the cable length increases, its insulation resistance decreases and vice-versa.



Insulation Resistance of a Single-Core Cable

Example 1: A single-core cable has a conductor diameter of 1cm and insulation thickness of 0.4 cm. If the specific resistance of insulation is $5 \times 10^{14} \Omega\text{-cm}$, calculate the insulation resistance for a 2 km length of the cable.

Solution 1:

Conductor radius,	$r_1 = 1/2 = 0.5 \text{ cm}$
Length of cable,	$l = 2 \text{ km} = 2000 \text{ m}$
Resistivity of insulation,	$\rho = 5 \times 10^{14} \Omega\text{-cm} = 5 \times 10^{12} \Omega\text{-m}$
Internal sheath radius,	$r_2 = 0.5 + 0.4 = 0.9 \text{ cm}$

$$\begin{aligned} R &= \frac{\rho}{2\pi l} \log_e \frac{r_2}{r_1} = \frac{5 \times 10^{12}}{2\pi \times 2000} \log_e \frac{0.9}{0.5} \\ &= 0.234 \times 10^9 \Omega = \mathbf{234 \text{ M}\Omega} \end{aligned}$$

Insulation Resistance of a Single-Core Cable

Example 2

The insulation resistance of a single-core cable is 495 MΩ per km. If the core diameter is 2.5 cm and resistivity of insulation is $4.5 \times 10^{14} \Omega\text{-cm}$, find the insulation thickness.

Solution.

Length of cable, $l = 1 \text{ km} = 1000 \text{ m}$

Cable insulation resistance, $R = 495 \text{ M}\Omega = 495 \times 10^6 \Omega$

Conductor radius, $r_1 = 2.5/2 = 1.25 \text{ cm}$

Resistivity of insulation, $\rho = 4.5 \times 10^{14} \Omega\text{-cm} = 4.5 \times 10^{12} \Omega\text{m}$

Let r_2 cm be the internal sheath radius.

$$\text{Now,} \quad R = \frac{\rho}{2\pi l} \log_e \frac{r_2}{r_1}$$

$$\text{or} \quad \log_e \frac{r_2}{r_1} = \frac{2\pi l R}{\rho} = \frac{2\pi \times 1000 \times 495 \times 10^6}{4.5 \times 10^{12}} = 0.69$$

$$\text{or} \quad 2.3 \log_{10} r_2/r_1 = 0.69$$

$$\text{or} \quad r_2/r_1 = \text{Antilog } 0.69/2.3 = 2$$

$$\text{or} \quad r_2 = 2 r_1 = 2 \times 1.25 = 2.5 \text{ cm}$$

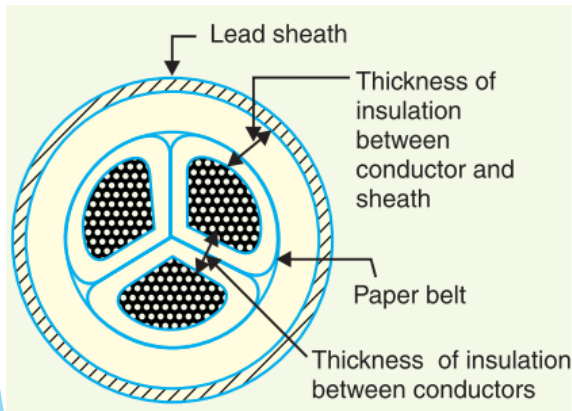
$$\therefore \text{ Insulation thickness} = r_2 - r_1 = 2.5 - 1.25 = \mathbf{1.25 \text{ cm}}$$

Please check the tutorial problems 1.2.3 from Mehta, Chapter 11, Page 274.

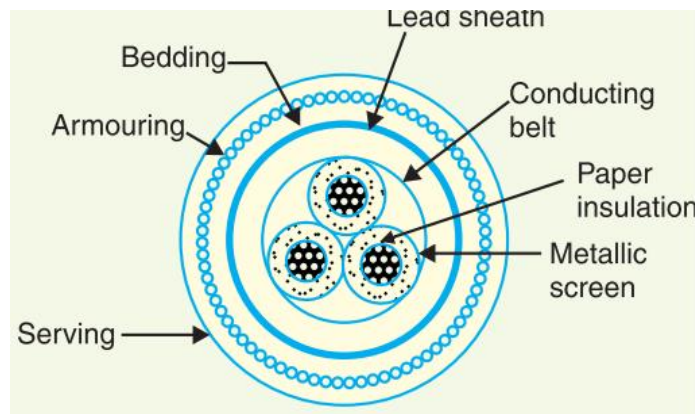
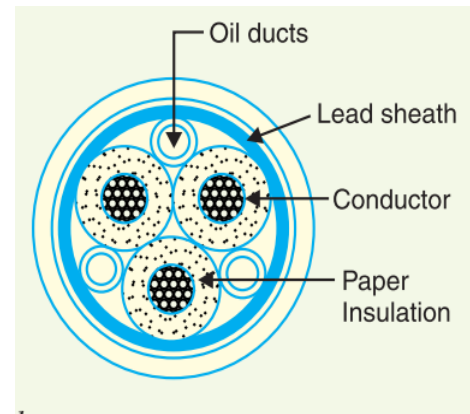
Cables for 3-Phase service

In practice, underground cables are generally required to deliver 3-phase power. For the purpose, either three-core cable or *three single core cables may be used. For voltages upto 66 kV, 3-core cable (*i.e.*, multi-core construction) is preferred due to economic reasons. However, for voltages beyond 66 kV, 3-core-cables become too large and unwieldy and, therefore, single-core cables are used. The following types of cables are generally used for 3-phase service :

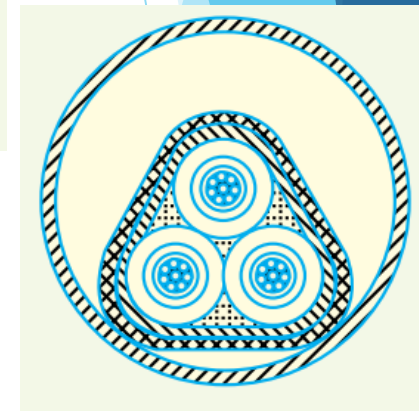
1. Belted cables — upto 11 kV
2. Screened cables — from 22 kV to 66 kV
3. Pressure cables — beyond 66 kV.



Belted Cable



Screened Cable



Pressure Cable

Cables for 3-Phase service

Feature

Operating Voltage

Belted Cable

Up to 11 kV ✗

1. Three cores are grouped together and insulated with a paper belt.
2. The gaps between the cores and the paper belt are filled with a fibrous material.
3. An overall lead sheath covers the belt.

Screened Cable

22 kV to 66 kV ✗

1. Each core is individually insulated and covered with a metallic screen (either foil or braid)
2. The three screened cores are then covered by a conducting belt and a lead sheath.

Pressure Cable

Above 66 kV ✗

1. The cable is placed in an airtight steel pipe or an overall metal sheath.
2. This enclosure is filled with either a pressurized oil or an inert gas (like nitrogen).

Construction

Advantages

1. Simple and low-cost construction.
2. Suitable for low-voltage applications.

1. Improved heat dissipation due to screens.
2. Higher current-carrying capacity.

1. Eliminates voids completely.
2. Higher operating temperature and current capacity.
3. Smaller cable size for the same voltage.
4. Defects can be easily detected by oil/gas leakage.

Reference:

Principles of Power System by V.K. Mehta and Rohit Mehta



QUESTIONS AND DISCUSSIONS

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